Dariel Angel Arro Moreno

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Professional summary

Software engineering graduate with hands-on experience building and evaluating back-end, full-stack, infrastructure, and usability-focused systems across C#, .NET, Python, JavaScript, TypeScript, Vue.js, Django, Docker, Linux, and relational databases. Brings practical strength in REST API design, clean architecture, repository pattern implementation, query optimization, observability, and codebase evaluation, aligned with work that requires understanding unfamiliar repositories, tracing system behavior, debugging, and reviewing maintainable solutions.

Education

UNIVERSIDAD DE LAS CIENCIAS INFORMÁTICAS (UCI)

Ingeniería en Ciencias Informáticas 2025

- Completed Computer Science Engineering studies with emphasis on software engineering and advanced algorithms.
- Built algorithmic problem-solving foundations through institutional representation in ICPC competitions, including exposure to graph theory, dynamic programming, segment trees, and heavy light decomposition.

Experience

UNIVERSIDAD DE LAS CIENCIAS INFORMÁTICAS

Backend Developer - TaskMaster API Jan 2026 - Present

- Built a RESTful API using clean architecture to separate domain, application, and infrastructure concerns for maintainable back-end development.
- Implemented relational persistence with SQLite and Entity Framework Core, applying the Repository Pattern to standardize data access.
- Applied modular design patterns to support clearer code organization and easier modification within the API codebase.

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Full-Stack Deployment and Monitoring Engineer Jan 2026 - Present

- Orchestrated a multi-container deployment environment on Ubuntu Server using Docker Compose for full-stack service coordination.
- Configured static network interfaces with Netplan to support stable server connectivity across containerized services.
- Implemented observability with Prometheus and Grafana, improving the ability to diagnose incidents and trace system behavior.

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Technical Evaluator - HCI Jan 2026 - Present

- Audited the usability of institutional platforms including XEJEL and Moodle LMS through structured Human Computer Interaction evaluation.
- Produced heuristic evaluation reports focused on reducing user cognitive load and identifying interface issues affecting platform usability.

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Dashboard SIPAC 4.0 Developer Jan 2024 - Dec 2025

- Designed and developed a full-stack institutional activity planning and control platform as a thesis project.
- Integrated Vue.js 3 front-end components with Django RESTful APIs to connect user workflows with back-end services.
- Optimized PostgreSQL queries to improve response times and strengthen database-backed application performance.
- Worked across front-end, back-end, and database layers, building practical experience tracing data through APIs, interfaces, and relational persistence.

Projects

PERSONAL PROJECT - github.com/DarielAngel/ciudad-bdi

Ciudad BDI - Agent-Based City Simulation Oct 2026

- Built a browser-based city simulation where 300 residents act as BDI (belief-desire-intention) agents with a deliberative layer and a reactive layer, so rush-hour traffic, late arrivals, and detours emerge from their decisions.
- Implemented A* routing over live traffic with a congestion cost, per-agent learning of commute times, and road closures that agents discover, remember, and route around.
- Designed a deterministic simulation engine decoupled from rendering (seeded random numbers, command queue, no DOM access) in dependency-free JavaScript modules that run as a static site.